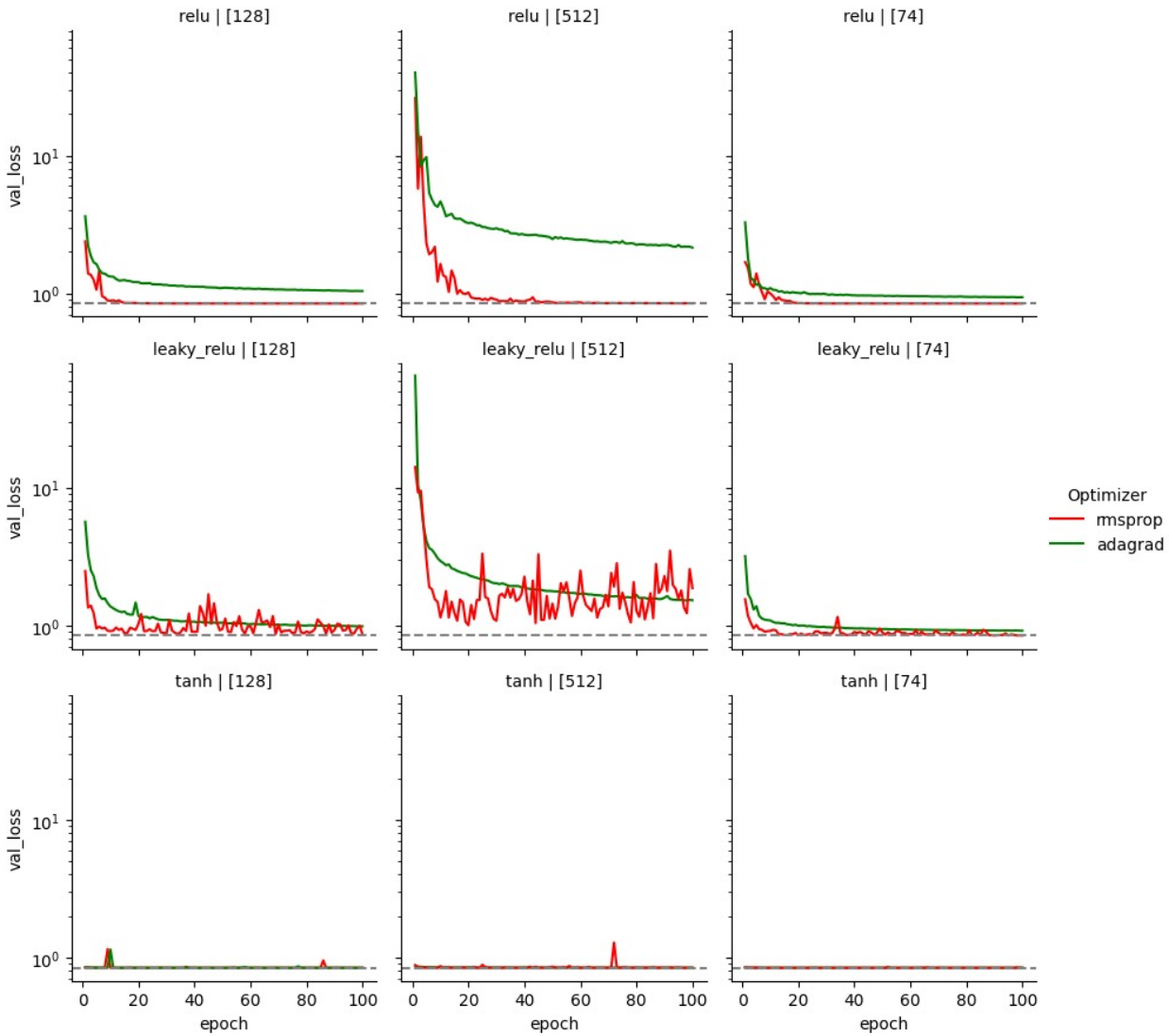
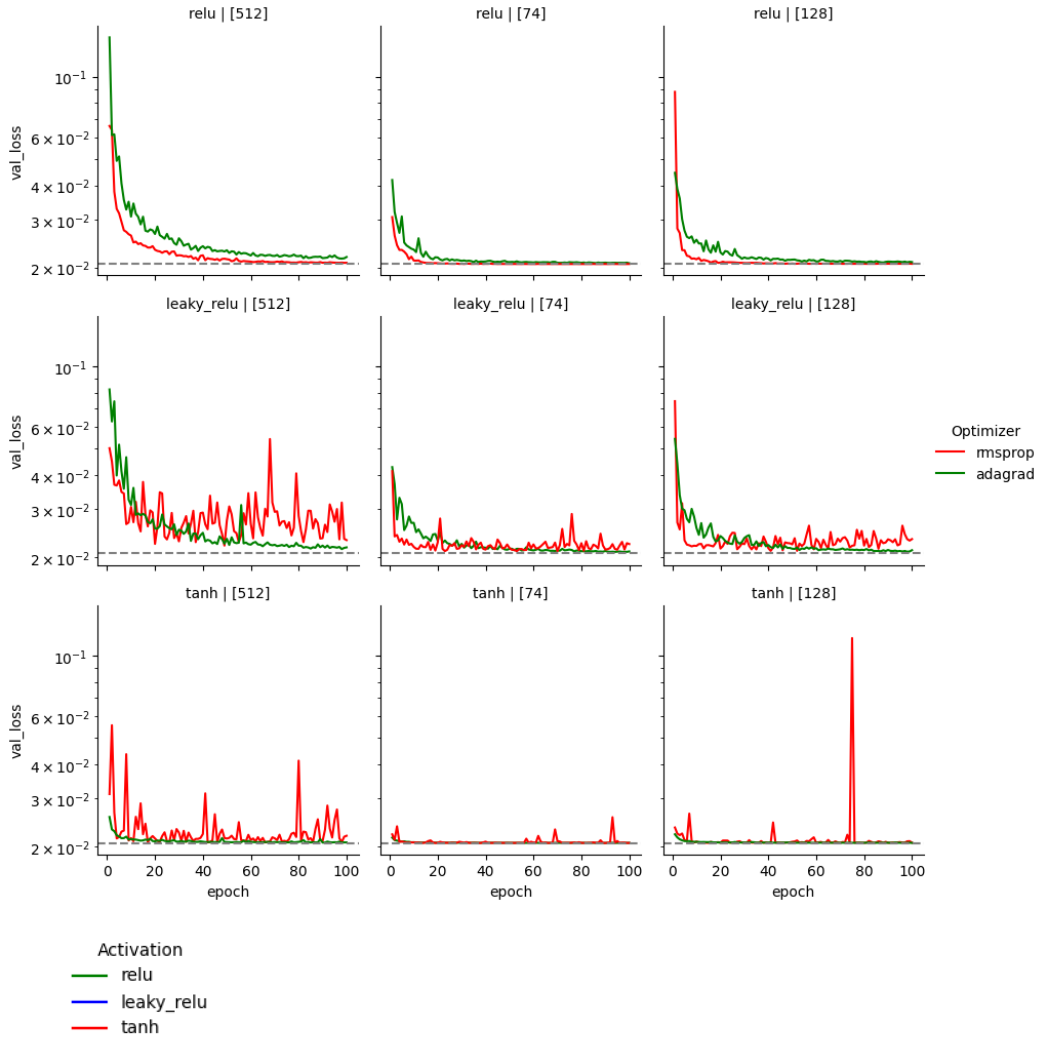
Network Structure

- Single layer with different hidden layer sizes
- MSE loss (unintentional, but results are same with SmoothL1)
- AdaGrad is not necessarily better than RMSProp always. But it is always more stable.
- Tanh converges in a few iterations

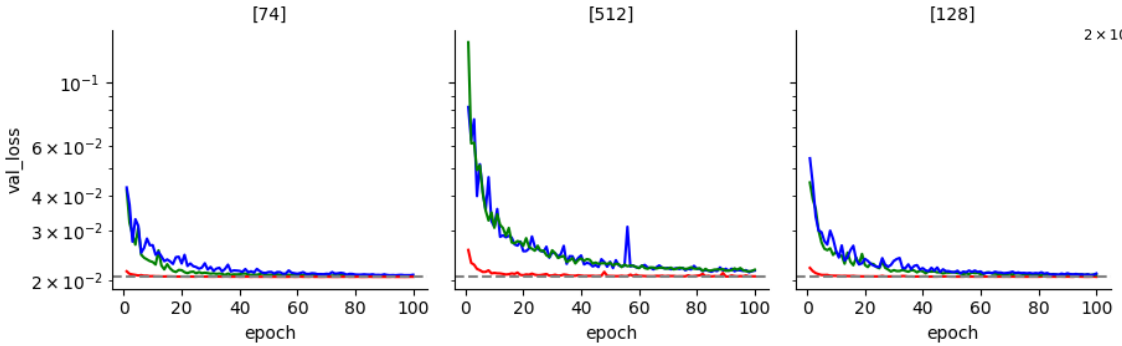


Same results with SmoothL1

- AdaGrad is good
- Tanh converges in few iterations
- Leaky ReLU isn't better than ReLU



Optimizer: AdaGrad, Loss Fn: SmoothL1

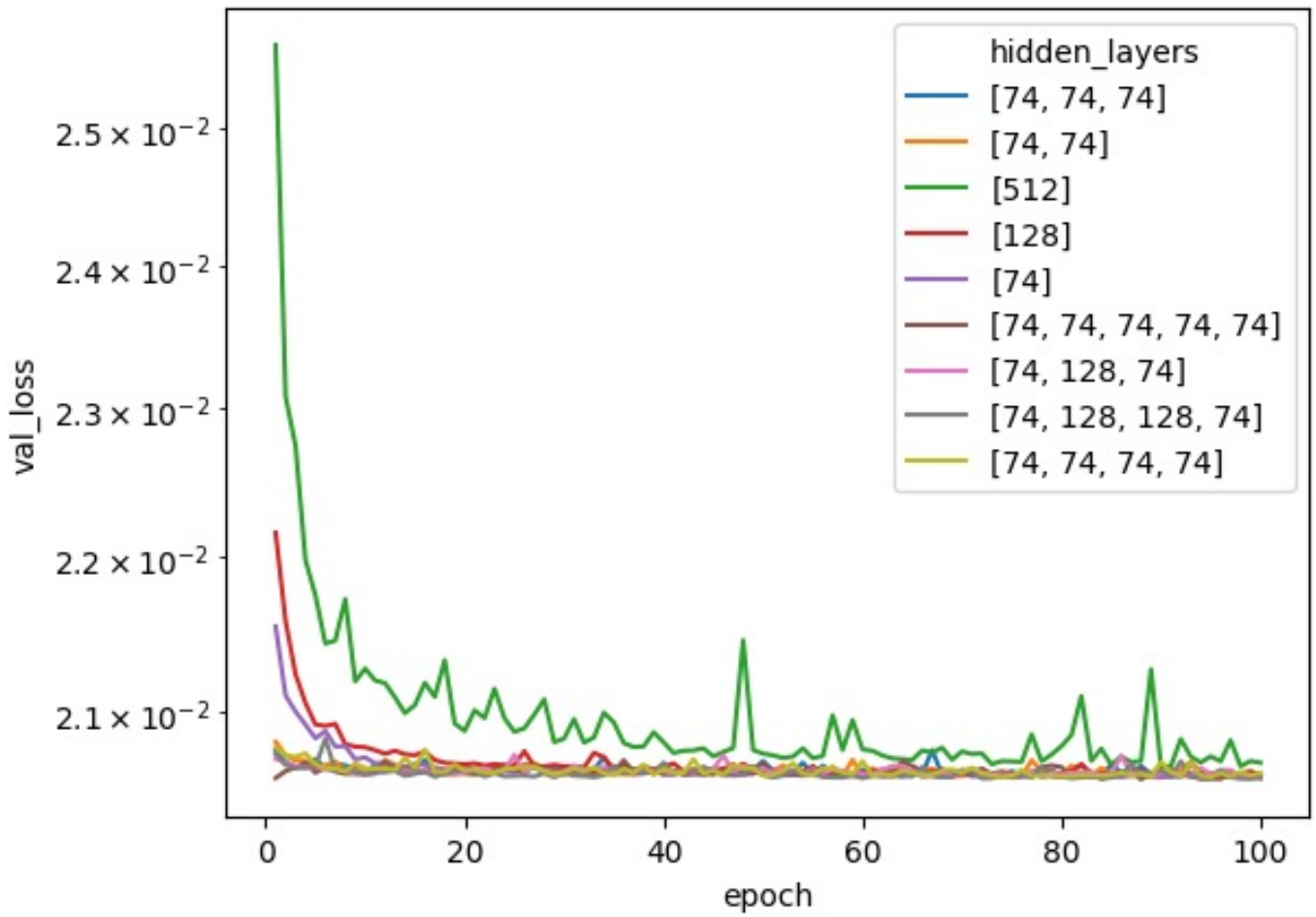


Comparision with different depths

- Not much improvement

(tanh, AdaGrad, smoothl1)

hidden_layers	
[74, 74, 74, 74, 74]	0.020587
[74, 128, 128, 74]	0.020591
[74, 128, 74]	0.020600
[74, 74]	0.020602
[74, 74, 74]	0.020603
[74]	0.020605
[128]	0.020606
[74, 74, 74, 74]	0.020620
[512]	0.020683



A network that always gives 0 as output performs the same.

```
SmoothL1 0.020573948091192687
L1 0.05524503815555577
L2 0.8433963852322892
```

The best trained network so far has 0.0205738 SmoothL1 loss.

This explains why u[0.1, 0.1] initialization, tanh activation gave good results.